

Instructor Answer Key

25 Code-Based Multi-Select MCQs from Lab Sheets 06–08

Each item has exactly two or three correct options. The explanations focus on what the code does and why the selected alternatives are valid.

Question 01: A, B

NumPy broadcasts the scalar intercept and applies the slope elementwise, so the output has one fitted value per input. The function performs regression prediction and does not modify `X` in place.

Question 02: C, D, E

Residuals are defined as observed minus predicted. Their squares are nonnegative, and `np.mean` produces the scalar MSE; no square root or absolute-value operation is used.

Question 03: A, C

The update subtracts a learning-rate-scaled gradient. The slope gradient explicitly includes `X`, and storing MSE over iterations reveals whether the loss is decreasing. Reversing the error sign also requires reversing the update sign.

Question 04: B, D, E

Scikit-learn expects a two-dimensional feature matrix, so the reshape produces shape $(n, 1)$. The intercept and coefficient are stored separately, and a new one-feature case must also be supplied in two-dimensional form.

Question 05: A, D

For a linear SVC, $Xw + b$ reconstructs the decision function. Its sign determines the class, but the raw score is not a calibrated probability. `support_` contains indices; `support_vectors_` contains coordinates.

Question 06: A, B, C

The expression directly implements $\max(0, 1 - y_i f(x_i))$ elementwise. Points safely beyond the margin have zero loss, while inside-margin and misclassified points have positive loss.

Question 07: A, E

SVR is a regression estimator and returns continuous predictions. With a linear kernel, its stored coefficient and intercept describe the fitted line, while `epsilon` defines the no-penalty tube.

Question 08: B, C, E

The tube is built by adding and subtracting epsilon. Strictly inside points have absolute error below epsilon and therefore zero loss. Only excess error beyond epsilon is retained.

Question 09: B, C

`fit_transform` estimates scaling parameters only from the training predictors. `transform` then reuses those parameters on the test predictors, avoiding test-information leakage.

Question 10: A, D, E

The fixed seed makes the random split reproducible, and stratification approximately preserves the target-class proportions. The four outputs separate development data from final evaluation data.

Question 11: B, D

`n_estimators` sets the tree count, `max_depth` limits tree complexity, and `max_features='sqrt'` randomizes the candidate predictors at splits. Random Forest trees are not a residual-correction sequence.

Question 12: A, C, E

`argsort` gives ascending positions and `[: :-1]` reverses them. Applying those positions to feature names preserves alignment. The importances are nonnegative model summaries, not causal effects or signed coefficients.

Question 13: B, E

The Cartesian product contains eight combinations, and five-fold cross-validation evaluates each five times, giving 40 validation fits. `best_params_` stores settings, not fitted coefficients.

Question 14: A, C, D

Gradient Boosting builds a correction sequence. The learning rate scales each new contribution and tree depth controls learner complexity; larger settings can overfit and are not guaranteed to generalize better.

Question 15: C, D

`predict` supplies class labels, and accuracy compares each test label with its corresponding prediction. Probability estimates would normally come from `predict_proba`, while regression error uses different metrics.

Question 16: A, B, E

The hidden labels are returned only because the synthetic generator knows the construction; the lab does not use them to train clustering. Standardization preserves the matrix shape while rescaling each feature column.

Question 17: C, E

Sampling indices without replacement selects distinct rows as initial centroids. The generator seed makes that random choice reproducible; no assignments or iterative updates have yet occurred.

Question 18: B, C, D

Broadcasting forms every observation-centroid difference, giving an $n \times K$ distance matrix. The minimum position selects a cluster, and the assigned rows are averaged to obtain updated centroids.

Question 19: D, E

Indexing `centroids[labels]` selects the assigned centroid for every row; the squared differences are then summed. Scikit-learn reports the same type of objective as `inertia_`, although label numbers may be permuted.

Question 20: A, B, D

`fit_predict` fits the model and returns one ID per row. There are three centroid rows and inertia is the fitted WCSS. Multiple initializations are tried, and cluster ID numbers are arbitrary.

Question 21: A, B

The linkage matrix stores the hierarchy created from the features. Complete linkage is based on the farthest cross-cluster observations, and a dendrogram can be built before choosing a horizontal cut or final cluster count.

Question 22: B, C

Agglomerative clustering produces assignments by merging observations according to the chosen linkage. It does not define mean centroids like K-Means or a noise label like DBSCAN.

Question 23: C, D

Scikit-learn uses `-1` for noise. The set of distinct labels therefore needs a one-label correction when noise exists, while direct comparison with `-1` counts the noise observations.

Question 24: D, E

DBSCAN can identify density-connected shapes and mark noise without a specified cluster count. K-Means is forced to return two centroid-based groups, so plotting both results exposes their structural differences.

Question 25: A, E

The scaler and model are fitted only from training data. The untouched test data is transformed using training statistics and then used for evaluation. The meaning of `score` depends on the estimator, and reproducibility does not guarantee quality.

Balanced key check: 15 questions contain two correct options and 10 contain three. Each answer position A–E occurs exactly 12 times.